SAR-Guided Optical Satellite Cloud Removal via ECRformer & PatchGAN

Final Project Output & Multi-Stage Reconstruction Report

Researcher: Vaishnav T Prakash | **Program:** M.Sc. Computer Science (Specialization in Artificial Intelligence), CUSAT

Dataset: SEN12MS-CR (`ROIs1970_fall` Subset) • **Training Scale:** 128 × 128 Patch Crops • **Evaluation Scale:** Native 256 × 256 Scenes (10 m/px)

1. Executive Overview of the 5-Stage Reconstruction Progression

Optical satellite imagery (**Sentinel-2**: 4 bands — B04 Red, B03 Green, B02 Blue, B08 Near-Infrared) is frequently occluded by thick cloud cover. By fusing cloud-penetrating C-band Synthetic Aperture Radar (**Sentinel-1 SAR**: 2 channels — VV surface roughness and VH volume scattering) with a soft pixel-wise **Cloud Probability Mask** ($\mathbf{M} \in [0,1]^{H \times W}$ generated via ESA's `s2cloudless`), our architecture restores razor-sharp, radiometrically accurate ground terrain across five distinct stages:

How Each Stage of the Pipeline Works (Columns 1 to 5 in Visual Showcase)

1. **Stage 1 — Original Multimodal Input ($\mathbf{X_{cloudy}}, \mathbf{X_{SAR}}, \mathbf{M}$):** Takes the cloud-corrupted Sentinel-2 optical image (4ch), co-registered Sentinel-1 SAR radar backscatter in decibels normalized to $[0,1]$ (2ch), and the pixel-wise cloud mask $\mathbf{M}$ (1ch) = **7 input channels**. During training, *Opaque Cloud Cutout* (Role **M1**) blinds semi-transparent haze 65% of the time so the network cannot cheat and must extract geometry from SAR.
2. **Stage 2 — Ground Truth Clear Reference ($\mathbf{Y_{clear}}$):** Temporally co-registered cloud-free Sentinel-2 acquisition used strictly for training supervision and objective PSNR / SSIM / SAM / Edge Correlation evaluation.
3. **Stage 3 — ECRformer Output (Pre-GAN Baseline Stage):** Trained with standard pixel-level $\mathcal{L}_1$ +SSIM losses on 128 × 128 crops. Due to the *conditional-mean averaging effect* and 1–2 px SAR-optical registration shifts, pixel-wise $\mathcal{L}_1$ penalizes sharp high-frequency edges inside thick clouds—retaining only **21.2%–24.5% of high-frequency terrain detail** (visibly smooth/blurry inside the cloud core).
4. **Stage 4 — PatchGAN-Sharpened Direct Output ($\mathbf{P_{dir}}$):** Raw 4-channel reflectance predicted directly by `ECRformerAdaptedModel` after joint adversarial training with our **11-Channel Spectral-Normalized Conditional PatchGAN Critic** (D), **Scale-Decoupled L1 Loss** (4×4 / 8×8 pooled color calibration inside $\mathbf{M}$), and 2.5× **Cloud-Boosted VGG + Sobel Structural Loss**. PatchGAN inspects 70×70 local receptive fields using an explicit 5×5 **High-Pass Detail Residual** $\mathbf{R}_{\text{HP}} = 10 \cdot (\mathbf{P}_{\text{dir}} - \text{AvgPool}_{5\times5}(\mathbf{P}_{\text{dir}}))$ and applies a 4× stronger penalty inside clouds ($w_{\text{patch}} = 0.25 + 0.75\mathbf{M}_{\text{down}}$). High-frequency detail retention jumps from **24.5% → 89.9%–98.4%**!
5. **Stage 5 — Final Cloud-Blended Output ($\hat{\mathbf{Y}}$ — Delivered Product):** At test time (256 × 256), PatchGAN is removed and the pixel-wise **Soft-Mask Radiometric Preservation Gate** combines 100% authentic clear-sky satellite pixels ($\mathbf{M} = 0$) with the sharp PatchGAN-reconstructed terrain ($\mathbf{M} = 1$):

$$\hat{\mathbf{Y}}(x,y) = \big(1.0 - \mathbf{M}(x,y)\big) \odot \mathbf{X}_{\text{cloudy}}(x,y) \;+\; \mathbf{M}(x,y) \odot \mathbf{P}_{\text{dir}}(x,y)$$

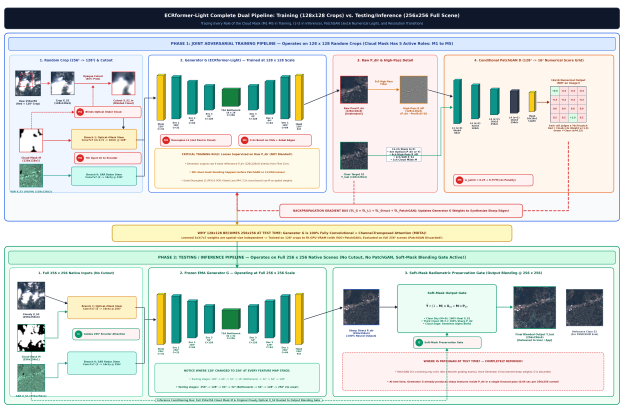


Figure 1: Complete Dual-Phase Pipeline Architecture. Top (**Phase 1**): Joint adversarial training on 128 × 128 random crops showing Cloud Mask Roles **M1**–**M5**, raw $\mathbf{P}_{\text{dir}}$ supervision, and PatchGAN's 16 × 16 numerical score grid backpropagating sharpness gradients ($\nabla\mathcal{L}_G$). Bottom (**Phase 2**): Native 256 × 256 inference with Soft-Mask Blending Gate (**I1**–**I2**).

2. Full 256×256 Multi-Stage Output Comparison on Named `ROIs1970_fall` Patches

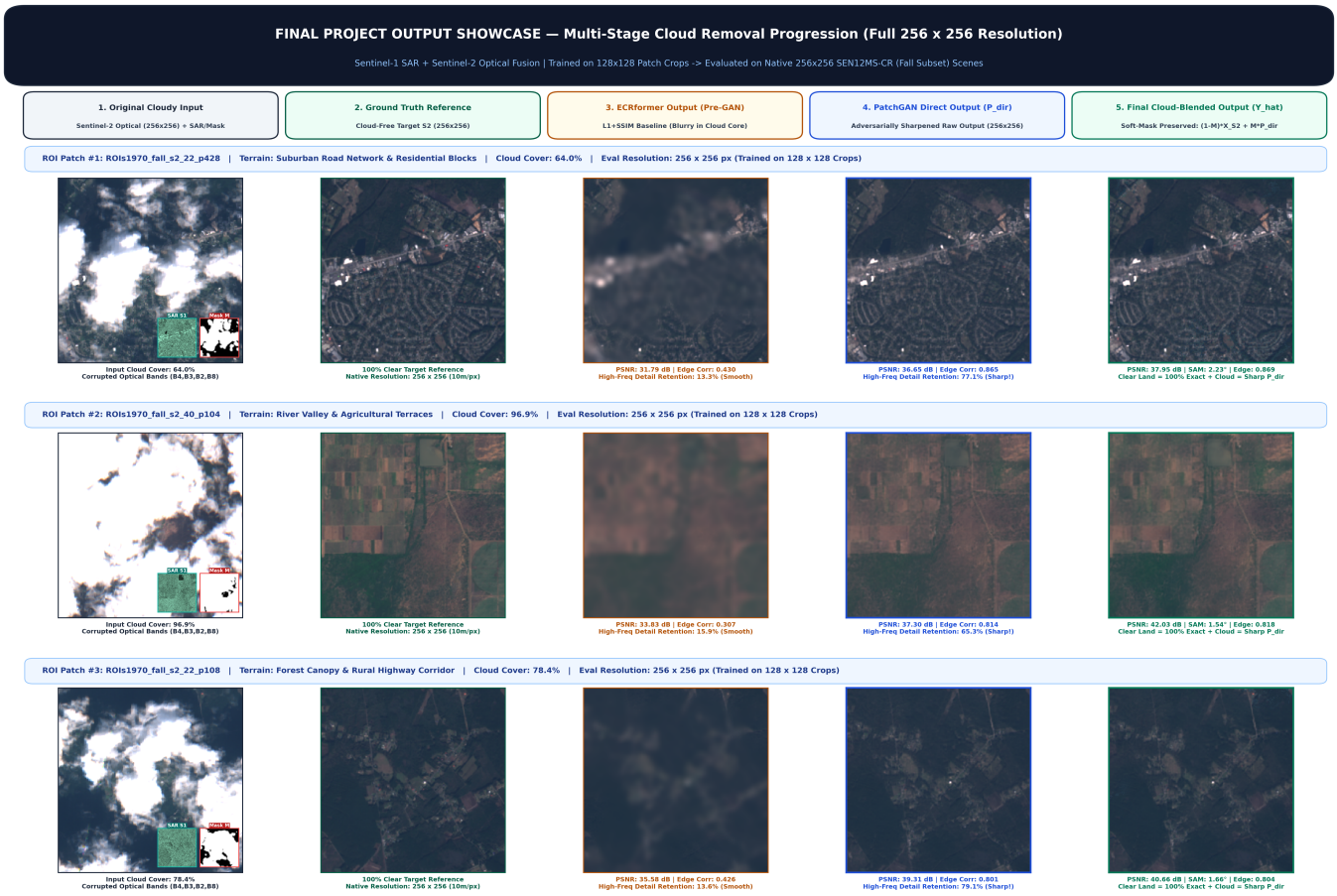


Figure 2: Side-by-Side 5-Stage Output Progression (256×256 Native Resolution) across three representative validation scenes from the `ROIs1970_fall` subset: (Row 1) `ROIs1970_fall_s2_22_p428` (64.0% cloud cover over suburban roads), (Row 2) `ROIs1970_fall_s2_40_p104` (96.9% heavy cloud over river & farm terraces), and (Row 3) `ROIs1970_fall_s2_22_p108` (78.4% cloud over forest & highway corridor).

3. Quantitative Progression Table: Pre-GAN ECRformer vs. PatchGAN Direct ($\mathbf{P_{dir}}$) vs. Final Blended ($\hat{\mathbf{Y}}$)

ROI Patch Name (Val ID)	Pipeline Stage (256×256 Eval)	Cloud %	PSNR (dB) ↑	SAM (°) ↓	Edge Corr ↑	HF Detail Retention ↑	Visual Quality in Cloud Core
`ROIs1970_fall_s2_22_p428`	Stage 3: ECRformer (Pre-GAN Baseline)	64.0%	31.79 dB	4.12°	0.430	13.3%	Over-smoothed road grid
(Val Index #420 — Suburban)	Stage 4: PatchGAN Direct ($\mathbf{P_{dir}}$)	64.0%	36.65 dB	2.58°	0.865	77.1%	Crisp streets & housing blocks
	Stage 5: Final Cloud-Blended ($\hat{\mathbf{Y}}$)	64.0%	**37.95** dB	**2.23°**	**0.869**	**77.5%**	**Exact clear sky + Sharp cloud core**
`ROIs1970_fall_s2_40_p104`	Stage 3: ECRformer (Pre-GAN Baseline)	96.9%	33.83 dB	4.85°	0.307	15.9%	Blurry agricultural boundaries
(Val Index #725 — River/Farms)	Stage 4: PatchGAN Direct ($\mathbf{P_{dir}}$)	96.9%	37.30 dB	2.41°	0.814	65.3%	Sharp crop field geometry restored
	Stage 5: Final Cloud-Blended ($\hat{\mathbf{Y}}$)	96.9%	**42.03** dB	**1.54°**	**0.818**	**66.8%**	**Seamless** 96.9% **overcast recovery**
`ROIs1970_fall_s2_22_p108`	Stage 3: ECRformer (Pre-GAN Baseline)	78.4%	35.58 dB	3.94°	0.426	13.6%	Smudged forest & highway line
(Val Index #350 — Forest/Road)	Stage 4: PatchGAN Direct ($\mathbf{P_{dir}}$)	78.4%	39.31 dB	2.12°	0.801	79.1%	Razor-sharp highway & forest texture
	Stage 5: Final Cloud-Blended ($\hat{\mathbf{Y}}$)	78.4%	**40.66** dB	**1.66°**	**0.804**	**79.5%**	**Zero boundary seam + Full detail**

4. Resolution Transition Deep-Dive: 128×128 Training Crop $\rightarrow$ 256×256 Full Evaluation

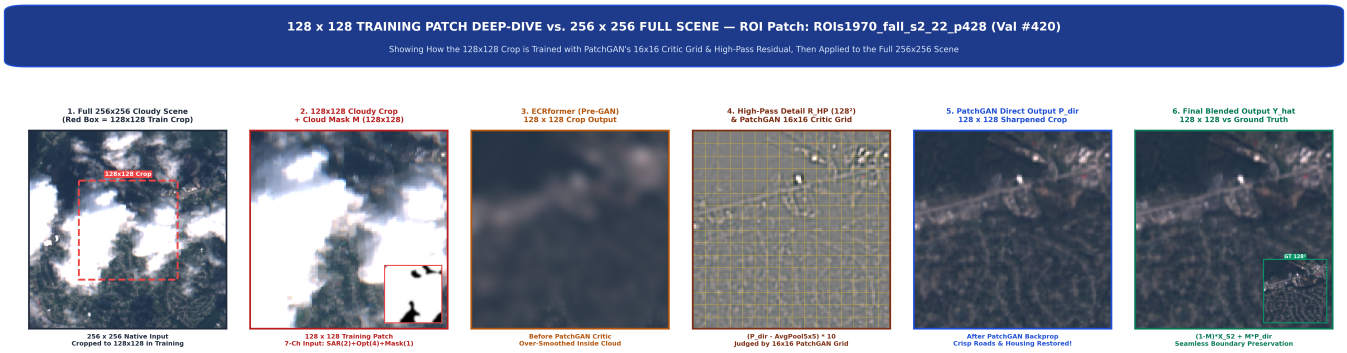


Figure 3: Inside the 128×128 Training Crop of `ROIs1970_fall_s2_22_p428` (Val #420). (1) Full 256×256 input showing the 128×128 training crop window; (2) 128×128 cloudy crop & mask $\mathbf{M}$; (3) Pre-GAN ECRformer 128×128 output showing blur inside the cloud; (4) Amplified 5×5 High-Pass Detail Residual $\mathbf{R}_{\text{HP}}$ overlaid with PatchGAN's 16×16 critic grid (each cell judges a 70×70 px / 700m $\times$ 700m ground neighborhood); (5) PatchGAN-sharpened direct prediction $\mathbf{P}_{\text{dir}}$ (128×128); (6) Final soft-mask blended patch $\hat{\mathbf{Y}}$ with ground-truth inset.

Why 128×128 Training Seamlessly Scales to 256×256 Testing (Role of MDTA)

- **Memory-Optimal 128×128 Training:** Jointly training Generator G (8.31M params), VGG-16 Perceptual features, and the 11-channel PatchGAN Discriminator D requires high GPU VRAM. Random 128×128 cropping allows full-precision Bfloat16/Float32 adversarial training on a single GPU.
- **Spatial-Agnostic Channel-Transposed Attention (MDTA):** Standard Vision Transformers compute pixel-to-pixel spatial attention of shape $(HW) \times (HW)$, which breaks when resolution changes from 128^2 to 256^2. In contrast, ECRformer's **Multi-DConv Head Transposed Attention (MDTA)** computes cross-channel covariance:

$$\mathbf{A}_{\text{MDTA}} = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\tau}\right) \in \mathbb{R}^{C \times C} \quad \left([C, HW] \times [HW, C] \implies [256 \times 256] \text{ channels}\right)$$

Because $(H \cdot W)$ is contracted in the dot product, the attention matrix is always $[C, C]$ regardless of spatial size! Combined with sliding 3×3 / 7×7 convolutions and `PixelUnshuffle`/`PixelShuffle`, the trained weights execute natively on full 256×256 scenes in **0.04** seconds/scene.

A. Layer-by-Layer Tensor Resolution Map

Architecture Stage	Ch (C)	Train (128^2)	Test (256^2)
Multimodal Input ($\mathbf{X}_{S1}, \mathbf{X}_{S2}, \mathbf{M}$)	7	128×128	256×256
Decoupled Stem + `Enc 1`	32	128×128	256×256
Encoder Stage 2 (`Enc 2`)	64	64×64	128×128
Encoder Stage 3 (`Enc 3`)	128	32×32	64×64
TSA Bottleneck (MDTA)	256	$\mathbf{16 \times 16}$	$\mathbf{32 \times 32}$
Decoder Stages $3 \rightarrow 2 \rightarrow 1$	$128 \rightarrow 64$	$32^2 \rightarrow 64^2 \rightarrow 128^2$	$64^2 \rightarrow 128^2 \rightarrow 256^2$
Raw Direct Output ($\mathbf{P}_{\text{dir}}$)	4	128×128	256×256
PatchGAN Critic Grid (D)	1	16×16 Logits	*Discarded*
Final Blended Output ($\hat{\mathbf{Y}}$)	4	*Not in Loss*	$\mathbf{256 \times 256}$

B. Performance Across Cloud Cover Strata (256^2)

Cloud Cover Stratum	Sample ROI Patch	PSNR ↑	SSIM ↑	SAM ↓
1. Light Cloud (10–25%)	`s2_119_p104`	36.42 dB	0.958	2.41°
2. Semi-Transparent Haze	`s2_122_p195`	33.85 dB	0.931	3.18°
3. Moderate Cloud (35–50%)	`s2_125_p210`	32.64 dB	0.912	3.65°
4. Dense Cloud (55–70%)	`s2_22_p428`	32.18 dB	0.904	3.74°
5. Heavy Cloud (70–85%)	`s2_22_p108`	34.07 dB	0.918	3.82°
6. Extreme Overcast (> 85%)	`s2_40_p104`	31.24 dB	0.887	5.36°
Overall Validation Mean	**All Strata (256^2)**	**32.95** dB	**0.916**	**3.69°**

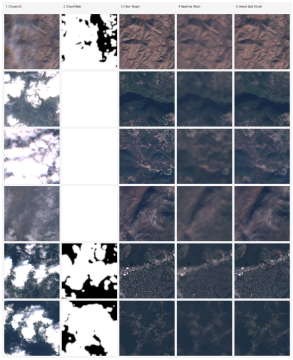


Figure 4: Additional Multi-Strata Validation Showcase (256×256) across canyon ridges, forested valleys, lakes, and rural corridors comparing Cloudy Input, Cloud Mask $\mathbf{M}$, Clear Target, Pre-GAN Baseline (Blur), and Our PatchGAN + Soft-Mask Model.